# FIITJEE

# ALL INDIA TEST SERIES

## CONCEPT RECAPITULATION TEST – III

## JEE (Main)-2021

**TEST DATE: 09-05-2021**

# ANSWERS, HINTS & SOLUTIONS

## *Physics*

**PART – A**

**SECTION – A**

1. B

Sol. $\frac{g}{4}=\frac{g}{\left(1+\frac{h}{R}\right)^2}\Rightarrow\left(1+\frac{h}{R}\right)^2=4$

$\Rightarrow 1+\frac{h}{R}=2\Rightarrow h=R$

2. B

Sol. $V=-\frac{GM}{R}$

$V'=-\frac{GM}{R/2}=\frac{-2GM}{R}$

$\therefore$ Gravitational Potential decreases.

3. D

Sol. $\frac{\mathrm{T}}{\mathrm{U}}=\frac{\frac{1}{2}m\omega^2\left(A^2-x^2\right)}{\frac{1}{2}m\omega^2x^2}=\frac{A^2-x^2}{x^2}$

4. D

Sol. Let initial conditions = $V,T$ and final conditions $V',T'$

$$\frac{V}{T}=\frac{V'}{T'}\Rightarrow\frac{V}{T}=\frac{V'}{1.2T'}\Rightarrow V'=1.2CV$$

So percentage of volume leaked out $=\frac{(1.2-0.9)V}{1.2V}\times100=25\%$

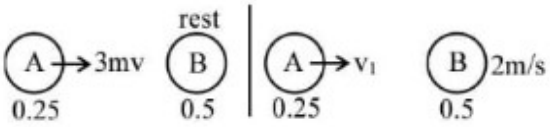
5. B

Sol. Let mass of boat = M; Mass of man = m displacement of boat

$$=\frac{m}{M+m}L=\frac{50}{450+50}\times10=1m$$

6. D

Sol. $mgh=\frac{1}{2}mv^2$




$$KE=\frac{1}{2}(0.5)v_B^2=1\Rightarrow3m/s$$

$$(0.25)/3=0.25v_i+0.5(2)$$

$$v_i=-1m/s$$

7. C

Sol. From Instantaneous axis of rotation $V_C>V_B>V_A$

8. A

Sol. With temperature rise (same 20°C for both), steel scale and copper wire both expand. Hence length of copper wire w.r.t. steel scale or apparent length of copper wire after rise in temperature

$$l_{app}=L'_{Cu}-L'_{steel}=\left[L_0(1+a_{Cu}\Delta\theta)-L_0(1+a_s\Delta\theta)\right]$$

$$=80\left(17\times10^{-6}-11\times10^{-6}\right)\times20$$

$$=0.0096cm$$

Length of the wire read $=80.0096\,\text{cm}$

9. B

Sol. $n_B=512\pm5=5017Hz\,or\,517\,Hz$ on loading was, $n_B\downarrow$, so it can't 507 otherwise 5 beat/sec will not be heard.

10. B

Sol. For max strain position of eye

$$M=\frac{f_0}{f_e}\left(1+\frac{f_e}{D}\right)$$

$$f_0=\frac{R}{2}=\frac{120}{2}=60\,cm$$

$f_e = 1.25cm$

$M = 50.4$

11. A

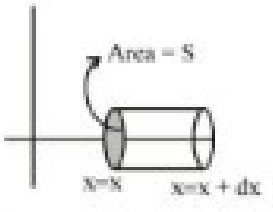
Sol. $\frac{1}{2}mv^2 + \frac{1}{2}I\frac{V^2}{R^2} = mgh$

$$V^2 = \sqrt{\frac{2gh}{1+\frac{I}{mR^2}}}$$

As I of solid sphere is least so it reach with maximum speed.

12. D

Sol. $E = -\frac{dv}{dx} = 3\alpha x^2$




$\phi_{in} = -\left(3\alpha x^2\right)s$

$\phi_{out} = \left\{3\alpha\left(x+dx\right)^2\right\}s = 3\alpha s\left(2xdx\right)$

$\phi_{net} = \left(6\alpha x\right)\left(sdx\right)$

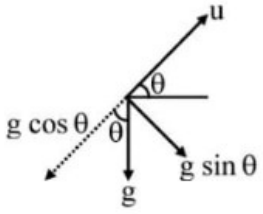
$\phi_{net} = \frac{q}{\varepsilon_0}\left(6\alpha x\right)\left(sdx\right)$

$\rho = \frac{q}{sd\alpha} = 6\alpha\varepsilon_0 x$

13. D

14. D

Sol. R.O.C. $= \frac{u^2}{g\sin\theta}$

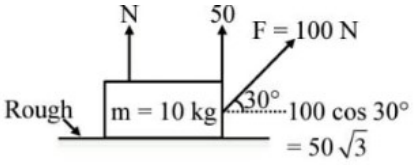



$a_t = g\cos\theta$

So, $\frac{a_t}{R.O.C.} = \frac{g\cos\theta}{u^2} \times g$

15. D

Sol. $N = 50, \qquad f = 50\sqrt{3}$




$R = \sqrt{50^2 + 3\times 50^2} = \sqrt{4\times 50^2}$

$= 2\times 50 = 25n$

$n = 4$

16. B

Sol. Torque $= \int_0^{\ell} Fx = \int_0^{\ell} F_0\left(1 - x/\ell\right)dx = F_0\ell - \frac{F_0}{\ell}\frac{\ell_0}{2} = \frac{F_0\ell}{2}$

17. B

Sol. $\overrightarrow{F_C} = -\left[\frac{\partial U}{\partial x}\hat{i} + \frac{\partial U}{\partial y}\hat{j} + \frac{\partial U}{\partial z}\hat{k}\right]$

Put $U = \frac{20yz}{x}$

$\overrightarrow{F_C} = \left(\frac{20yz}{x^2}\right)\hat{i} - \left(\frac{20z}{x}\right)\hat{j} - \left(\frac{20y}{x}\right)\hat{k}$

18. A

Sol. $\phi = Li$

$EMF = -\frac{d\phi}{dt} = -\frac{Ldi}{dt}$

If $i = i_0 \sin\left(2\pi f_1 t\right)$

$\frac{e_1}{e_2} = \frac{f_1}{f_2}$

19. A

Sol. $\sigma = en\left[\mu_e + \mu_h\right]$

20. A

Sol. $\mathrm{U}_{av} = \in_0 E^2_{rms}$

Total average energy density of electromagnetic wave

## SECTION – B

21. 00076.00

Sol. $V_1\rho_1 g + V_2\rho_2 g = mg.....(1)$

$\Rightarrow m = 40\times 1 + 60\times 0.6 = 40 + 36 = 76gm$

22. 00000.30

Sol. $y=\dfrac{T/A}{\Delta\ell/\ell}\Rightarrow\Delta\ell=\dfrac{T\ell}{Ay}=\dfrac{(10)(3)}{(10^{-6})(10\times10^{10})}$

$\Rightarrow\Delta\ell=3\times10^{-4}m=0.3\,mm$

23. 00002.00

Sol. $V_0=\sqrt{\dfrac{GM}{R}}$.....(1)

$V_e=\sqrt{\dfrac{G(8M)}{2R}}$.....(2)

Given $\dfrac{M}{4/3\pi R^3}=\dfrac{M'}{4/3\pi(8R^3)}\Rightarrow M'=8M$

$\Rightarrow V_e=2V_0\Rightarrow So, n=2$

24. 00002.00

Sol. $x=A\sin\omega t$

$\dfrac{A}{2}=A\sin\omega t\Rightarrow\sin\omega t=\dfrac{1}{2}\Rightarrow\omega t=\dfrac{\pi}{6}$

$\Rightarrow t=\dfrac{\pi}{6\omega}=\dfrac{\pi}{6(2\pi/T)}=\dfrac{T}{12}=\dfrac{24}{12}=2\sec$

25. 00410.00

26. 00010.00

Sol. $f=\dfrac{1}{2l}\sqrt{\dfrac{T}{m}}=\dfrac{1}{2l}\sqrt{\dfrac{T}{\pi r^2 d}}$

$\dfrac{f_1}{f_2}=\dfrac{l_2}{l_1}\sqrt{\dfrac{T_1}{T_2}\left(\dfrac{r_2}{r_1}\right)^2\left(\dfrac{d_2}{d_1}\right)}$

$=\dfrac{35}{36}\sqrt{\dfrac{8}{1}\left(\dfrac{1}{4}\right)^2\times\dfrac{2}{1}}$

$\dfrac{f_1}{360}=\dfrac{35}{36}\Rightarrow f_1=350\,Hz$

$\Rightarrow Beats=f_2-f_1=360-350=10\,Hz$

27. 00004.00

28. 00006.00

Sol. $mg+V\rho g=Mg$

$\Rightarrow m=M-V\rho$

For m to be minimum V should be maxima

$$\Rightarrow m = 10 - \frac{10}{2.5} = 6$$

29. 00016.00

Sol. Charge enclosed $q = \int_0^{R/2} Kr \; 4\pi r^2 dr = \frac{4\pi K}{4}\left(R/2\right)^4$

$$q = \frac{\pi K R^4}{16}$$

Using gauss law; $E4\pi\left(R/2\right)^2 = \frac{\pi K R^4}{16\varepsilon_0}$

$$\mathrm{E} = \frac{\mathrm{KR}^2}{16\varepsilon_0} \text{ so, } \mathrm{N} = 16$$

30. 00259.00

Sol. $f - 256 = x \; .....(1)$

$262 - 256 = 2x \; .....(2)$

$\Rightarrow f = 259\,Hz$

# Chemistry

## PART – B

### SECTION – A

31. C

Sol. At constant volume $\frac{p_1}{T_1}=\frac{p_2}{T_2}\Rightarrow p_2=p_1\frac{T_2}{T_1}=\frac{3}{2}$

$\Rightarrow\ dG=Vdp-SdT$

$=V\Delta p-\int_{200k}^{300k} 2+10^{-T}T\,dT$

32. D

Sol. $r_1=k\left[A\right]^n\left[B\right]^m$

$r_2=k\left[2A\right]^n\left[\frac{B}{2}\right]^m=2^{n-m}k\left[A\right]^n\left[B\right]^m$

$\frac{r_1}{r_2}=2^{n-m}$

33. C

Sol. When a reaction is reversed, magnitude of enthalpy change remains same.

34. B

Sol.




35. B

Sol. $p\alpha n$ at constant volume of temperature.

$2NO(g)+Cl_2(g)\rightleftharpoons 2NOCl(g)$

$2p \qquad p \qquad O$

$2p-2x \qquad p-x \qquad 2x$

$p_T=2p-2x+p-x+2x=1$

$3p-x=1\ \ \&\ 2x=\frac{p-x}{4}$

Solving $x=\frac{1}{26}\ \&\ p=\frac{9}{26}$

$k_p=\frac{2(x)^2}{(p-x)(2p-2x)^2}$

36. D

Sol. $E\alpha\frac{1}{\lambda}$

Maximum wavelength radiation means minimum energy radiation which is emitted when transition occurs from $4 \longrightarrow 2$.

$$\frac{1}{\lambda} = R\left(\frac{1}{4} - \frac{1}{16}\right)$$

37. B

Sol. Given silicate anion is $\left[\left(SiO_3\right)^{2-}\right]_6$

$\therefore n = 12$

38. C

Sol. Chemisorption increases with increase in temperature initially and then decrases.

39. A

Sol. $\left[I^-\right] = \sqrt{\frac{7.2\times10^{-9}}{0.005}} = 1.2\times10^{-3} M$

Maximum moles of NaI that can be added to 500 ml of $Pb\left(NO_3\right)_2$ are

$$\frac{1.2\times10^{-3}}{1000}\times500 = 6\times10^{-4} moles \Rightarrow 0.09g$$

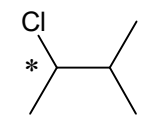
40. A

Sol. Equivalents of $Cr_2O_7^{2-} = 0.136$

$$n_b \times n = 0.136 \Rightarrow n = \frac{0.136}{6} = 0.226$$

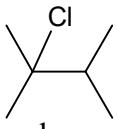
41. B

Sol.




two stereoisomers

$(R)\&(S)$

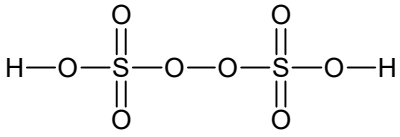



only one product

Total products = 3

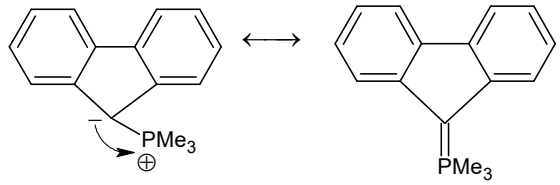
42. A

Sol. $H_2SO_4$ on electrolysis forms $H_2S_2O_8$



43. C

Sol.




Negative charge can be delocalised on phosphorous also in addition to phenyl rings. So the anion is more stabilised by resonance.

44. C

Sol. Atomic radii increases for $\mathrm{Eu}$ & $\mathrm{Yb}$ in lanthanoids series.

45. A

Sol. Epoxide undergo ring opening with bases. Greater the ring strain, greater is the reactivity towards base.

46. D

Sol. Protonated methanol is a stronger acid than methanol.

47. B

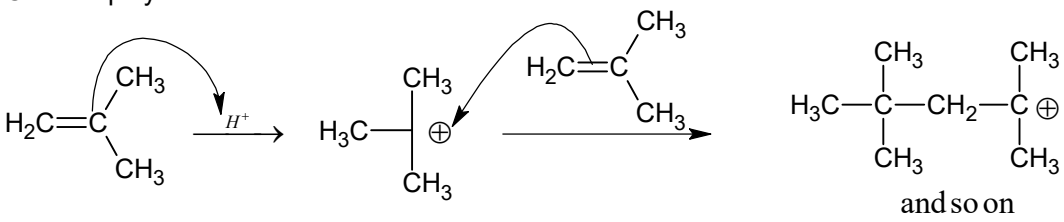
Sol. Reactivity of carbonyl compounds follow the order acid chlorides > aldehydes > ketones > ester

48. C

Sol. $\mathrm{OH}^-$ Converts phenol to phenoxide ion which is strongly activating so attack occurs at ortho position w.r.t. hydroxyl group

49. A

Sol. Cationic polymerisation occurs as:




and so on

50. C

Sol. $\left[Cr\left(NH_3\right)_6\right]^{3+}$ - $\left(\mathrm{d}^3\right)$-$\mathrm{d}^2\mathrm{sp}^3$ hybridised & paramagnetic

$\left[Co\left(CN\right)_6\right]^{3-}$ - $\left(\mathrm{d}^6\right)$-$\mathrm{d}^2\mathrm{sp}^3$ hybridised ($\mathrm{CN}^-$ is a strong field ligand)

## SECTION – B

51. 00009.00

Sol. $\mathrm{W}_7\mathrm{X}_6\mathrm{Y}_4$

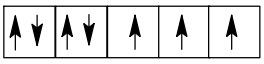
52. 00004.00

Sol. i ,ii, iii, iv are used

53. 00004.0

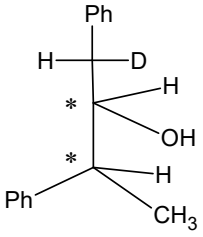
Sol. ii, iii, iv, vi

54. 00005.00

Sol. $Fe^{+} \rightarrow 3d^6 4s^1 \xrightarrow{pairing\ of\ one\ electron}$ 

$\left(t_{2g}{}^5\right)\left(e_g{}^2\right)$

55. 00004.00

Sol. 


56. 00015.38

Sol. $A \rightarrow B\ w_1 = 0,\ \Delta U_1 = 1 \times C_V(T), q_1 = C_V(T)$

$B \rightarrow C\ w_2 = -2p_0V_0,\ \Delta U_2 = 1 \times C_V(2T), q_2 = C_V(2T) + 2p_0V_0$

$C \rightarrow D\ w_3 = 0,\ \Delta U_3 = 1 \times C_V(-2T), q_3 = -C_V(2T)$

$D \rightarrow A\ w_4 = P_0V_0,\ \Delta U_4 = -1 \times C_V(T), q_4 = -C_V(T) - P_0V_0$

$$\text{Efficiency} = \frac{|W|}{q_{absorbed}} = \frac{P_0V_0}{C_VT + C_V2T + 2P_0V_0} = \frac{RT}{3 \times \frac{3}{2}R \times T + 2RT} = \frac{2}{13} = 0.1538$$

57. 00001.00

Sol. GALVG

58. 00003.00

Sol. $V \propto n$

$$\frac{\text{Anode}}{\text{Cathode}} = 1\text{mole}(C_2H_6) + 2\text{mole}(CO_2) / 1\text{mole}(H_2)$$

59. 00003.00

Sol. $2KI + HgI_2 \rightarrow K_2[HgI_4] \rightarrow 2K^{\oplus} + [HgI_4]^{2-}$

60. 00002.00

Sol. (342 – p + 352 – p + p = 440)

$\therefore$ p = 254

$$k_p = \frac{254}{88 \times 98}$$

$$k_p' = \frac{88 \times 08}{254}\text{mm Hg} = \frac{88 \times 98}{254} \times \frac{1}{760}\text{atm} = 0.04$$

$$0.04 = k_p = \frac{n_{CO}n_{Cl_2}}{n_{COCl_2}} \times \left(\frac{p}{m}\right)^{\Delta n} = \frac{\alpha^2}{1-\alpha} \times \frac{1\text{atm}}{1-\alpha}$$

$\therefore \alpha = 0.026$

x = 20.6 %

$\therefore$ x/10 = 2.06% $\approx$ 2%

# Mathematics

## PART – C

### SECTION – A

61. B

Sol. $\frac{12}{3}+\frac{6}{2}+\frac{15}{5}+\frac{4n}{3}=26 \Rightarrow n=12$

Required coefficient = ${}^{26}C_4 \times {}^{22}C_3 \times \left({}^{19}C_3 \times 2^3\right) \times {}^{16}C_{16}$.

62. A

Sol. $ax^2-(4-2a)x-8=0$ for $x=-2,\frac{4}{a}$

By the condition $1<\frac{4}{a}\le 2 \Rightarrow a \in [2,4)$.

63. D

Sol. n should not be divisible by the prime factors of 350
ie 2, 5 or 7
$(n(2)+n(5)+n(7))-(n(10)+n(14)+n(35))+n(70)=329$.

64. B

Sol. Either both quadratic expression should be perfect squares or both should have same roots.
$D_1 = 0$ and $D_2 = 0$ do not hold for same k.

Both equation are identical if $\frac{2}{1}=\frac{-4(k-1)}{2k}=\frac{3}{k+1}$ for $k=\frac{1}{2}$.

65. D

Sol. The lines should intersect or be parallel
Let $(2t+1,-t,2t)$ be any point on the first line

Putting it in both the planes we get $\lambda=\frac{-31}{7}$.

There is no $\lambda$ for which both the lines are parallel.

66. B

Sol. $n^2+n(a-1)-153=0$ where a is the first term.

No. of terms = No. of factors of 153 excluding 1 and 153.

67. D

Sol. Let foot of perpendicular from Q to the directrix be N.
$\Rightarrow$ SQ + PQ = QN + PQ is minimum if P, Q & N are collinear.
So minimum value of SQ + PQ = PN = 1 – b.

68. D

Sol. $f'_{n+1}(x)=f'_1(x)e^{f_1(x)}e^{f_2(x)}.....e^{f_n(x)}=f_1(x)f_2(x).....f_{n+1}(x)$

Putting *n* = 9 we get $f'_{10}(x)=\prod_{r=1}^{10} f_r(x)$.

69. B

Sol. The point of intersection $\left(at^2, 2at\right)$ lies on the curve $y^2 + x - y - 2 = 0$

$\therefore 4a^2t^2 + at^2 - 2at - 2 = 0$

$\left(4a^2 + a\right)t^2 - 2at - 2 = 0$

Sum of roots $t_1 + t_2 = \dfrac{2a}{4a^2 + a}$

Product of roots $t_1t_2 = -\dfrac{2}{4a^2 + a}$

Point of intersection of tangents $(x, y) = \left(at_1t_2, a\left(t_1 + t_2\right)\right)$

$x = -\dfrac{2a}{4a^2 + a}, y = \dfrac{2a^2}{4a^2 + a}$

$\dfrac{y}{x} = -a$

$\therefore x = -\dfrac{2\left(-\dfrac{y}{x}\right)}{4\left(\dfrac{y}{x}\right)^2 + \left(-\dfrac{y}{x}\right)}$

$x = \dfrac{2xy}{4y^2 - xy}$

$1 = \dfrac{2}{4y - x} \Rightarrow 4y - x = 2$ .

70. C

Sol. $BO \perp AC$

$\dfrac{y-7}{x-4} \cdot \dfrac{\alpha}{1-\alpha} = -1 \Rightarrow \dfrac{\alpha - 1}{\alpha} = \dfrac{y-7}{x-4}$

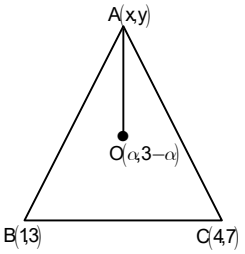
$1 - \dfrac{1}{\alpha} = \dfrac{y-7}{x-4} \Rightarrow \dfrac{1}{\alpha} = 1 - \dfrac{y-7}{x-4} \Rightarrow \alpha = \dfrac{x-4}{x-y+3}$

$AO \perp BC$

$\dfrac{y - 3 + \alpha}{x - \alpha} \times \dfrac{4}{3} = -1 \Rightarrow 4y - 12 + 4\alpha = -3x + 3\alpha$

$\alpha = -3x - 4y + 12$

$\therefore$ The locus is $\dfrac{x-4}{x-y+3} = -3x - 4y + 12$ .




71. C

Sol. Sum of 7 digits = a multiple of 9

Sum of number 1, 2, 3, 4, 5, 6, 7, 8, 9 = 45 so two left should also have sum of 9.

The pair of left numbers is (1, 8), (2, 7), (3, 6), (4, 5).

with each pair left number of 7 digit number = 7!

So with all 4 pairs = $4 \times 7!$ .

72. B

Sol. Let the probability of choosing integer k

p (k) = $\dfrac{\alpha}{k^4}$ , $\alpha$ is constant of proportionality

$\sum_{k=1}^{2m} p(k) = 1 \Rightarrow \alpha = \left(\sum_{k=1}^{2m} \frac{1}{k^4}\right)^{-1}$

a = p (choosing an odd number)

$a = \sum_{k=1}^{m} p(2k-1) = \alpha \cdot \sum_{k=1}^{m} \frac{1}{(2k-1)^4}$

$1 - a = \alpha \cdot \sum_{k=1}^{m} \left(\frac{1}{2k}\right)^4 < \alpha \cdot \sum_{k=1}^{m} \frac{1}{(2k-1)^4} = a$

$\therefore$ a > 1/2.

73. C

Sol. Total number of ways to place 11 identical balls $= {}^{13}C_2$

Total number of ways to place the balls so that the first box will have more balls than the second and third boxes combined= $= {}^{7}C_2$

Therefore, required cases = ${}^{13}C_2 - 3{}^{7}C_2$

74. D

Sol. $\begin{vmatrix} \lambda-1 & 3\lambda+1 & 2\lambda \\ \lambda-1 & 4\lambda-2 & \lambda+3 \\ 2 & 3\lambda+1 & 3(\lambda-1) \end{vmatrix} = 0 \Rightarrow \lambda = 0 \text{ or } 3.$

If $\lambda = 0,$ the equations become $-x + y = 0, -x - 2y + 3z = 0$ and $2x + y - 3z = 0,$

$\therefore \frac{x}{6-3} = \frac{y}{6-3} = \frac{z}{-1+4}$

75. A

76. A

Sol. There is no point of intersection.

77. B

78. C

Sol. a, b, c are in A.P $\Rightarrow$ 2b = a + c

$\text{SinB. SinC.Cos}^2\frac{A}{2} = \frac{b}{2R}.\frac{C}{2R}.\frac{S(S-a)}{bc} = \frac{S(S-a)}{4R^2}$ …(1)

$\text{Similarly SinA Cos}^2\frac{B}{2}\text{SinC} = \frac{S(S-b)}{4R^2}$ …(2)

$\text{SinA SinB Cos}^2\frac{C}{2} = \frac{S(S-c)}{4R^2}$ …(3)

(1) + (3) = $\frac{S(S-a)+S(S-c)}{4R^2} = \frac{S(2S-(a+c))}{4R^2} = \frac{S(2S-2b)}{4R^2}$ [$\because$ 2b = a + c]

$= \frac{2S(S-b)}{4R^2} = 2\ \text{SinA Cos}^2\frac{B}{2}\text{SinC}$

$\therefore \text{SinBSinCCos}^2\frac{A}{2}$, $\text{SinA Cos}^2\frac{B}{2}\text{SinC}$ and $\text{SinA SinB Cos}^2\frac{C}{2}$ are in A.P

79. D

Sol. The number of favorable cases is ${}^nC_0 \times 2^n + {}^n C_1.2^{n-1} + \ldots. + {}^n C_n\, 2 = 3^n$.

Total number of cases is $2^n \times 2^n = 4^n$. Thus required probability is $\left(\frac{3}{4}\right)^n$

80. A

Sol. Let $(\lambda, \lambda, \lambda)$ be the point of intersection of two lines.

$\Rightarrow \lambda(\sin A + \sin B + \sin C) = 2d^2$ and $\lambda(\sin 2A + \sin 2B + \sin 2C) = d^2$

$\Rightarrow \dfrac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C} = \dfrac{1}{2}$

$\Rightarrow \sin\dfrac{A}{2}\sin\dfrac{B}{2}\sin\dfrac{C}{2} = \dfrac{1}{16}$.

## SECTION – B

81. 00005.00

Sol. The system has a nonzero solution

$\therefore |A| = 0$

$$\begin{vmatrix} 2 & 3 & -1 \\ 3 & 2 & k \\ 4 & 1 & 1 \end{vmatrix} = 0$$

The solution is $(2a, -3a, -5a)$

Least positive integer value of $z$ is 5.

82. 00000.00

Sol. $f(x) = \int\limits_{\frac{\pi}{2}}^{x} (\cos t - 2\sin t)dt$

$f'(x) = \cos x - 2\sin x < 0$ in $\left[\dfrac{\pi}{2}, \pi\right]$

$\therefore f(x)$ is decreasing function

$\therefore$ Greatest value = $f\left(\dfrac{\pi}{2}\right) = 0$.

83. 00006.00

Sol. $f(x) = x\sin\left([x]^2 - 1\right)\left([x]^2 - 4\right)$

$\lim\limits_{x\to 0} f(x) = 0$

$\therefore f(x)$ is continuous at $x = 0$

$\lim\limits_{x\to 1^-} f(x) = -1\sin\left((-1)(-4)\right) = -\sin 4$

$\lim\limits_{x\to 1^+} f(x) = 1\sin 0 = 0$

$\therefore$ discontinuous at $x = 1$

$\lim\limits_{x\to 2^+} f(x) = 0 = \lim\limits_{x\to 2^-} f(x) = 0$

$\therefore$ continuous at $x = 2$

similarly it is continous at $x = -1$ but not at $-2$

$f(x)$ is discontinuous at $-4, -3, -2, 1, 3, 4$.

84. 00005.00

Sol. The plane passes through the point of intersection of lines $(0,1,1)$

The point $(2,3,2)$ lies on the line, whose projection on the second line is $(1,2,0)$

$\therefore$ the required plane has normal $(1,1,2)$

$\therefore$ Equation of the plane is $1(x-0)+1(y-1)+2(z-1)=0$

$x+y+2z-3=0$

$(k,-2,0)$ lies on the plane

$\therefore k=5$.

85. 00006.00

Sol. 6 boys and 5 girls can be arranged in 11! ways
but the 6 boys and 5 girls cannot arrange themselves

$\therefore$ No. of ways $=\dfrac{11!}{6!5!}$

The teacher are arranged in the remaining places in $12P_3$ ways.

$\therefore 12P_3 \times 11_{C_5}$

$12\times11\times10\times11_{C_5}$

$6\times220\times11_{C_5}$

$k=6$.

86. 00001.00

Sol. $dy = x\tan y(\sin y-1)dx$

$\dfrac{dy}{dx}+x\tan y = x\tan y.\sin y$

$\cos y\dfrac{dy}{dx}+x\sin y = x\sin^2 y$

Let $\sin y = t$

$\dfrac{dt}{dx}+xt=xt^2$, $\dfrac{1}{t^2}\dfrac{dt}{dx}+x\left(\dfrac{1}{t}\right)=x$

$\dfrac{1}{t}=u$

$=-\dfrac{du}{dx}+xu=x$

$\dfrac{du}{dx}+(-x)u=-x$

$ue^{-\int xdx}=\int -xe^{-\int xdx}dx$

$ue^{-\frac{x^2}{2}}=-\int xe^{-\frac{x^2}{2}}dx \;=e^{-\frac{x^2}{2}}+c$

$u=1+ce^{\frac{x^2}{2}}$

$t=\dfrac{1}{1+ce^{\frac{x^2}{2}}}$

$\sin y=\dfrac{1}{1+ce^{\frac{x^2}{2}}}$.

87. 00004.00

Sol. $x + y + z = 20$

No. of positive solution $19_{C_2}$

They form a triangle if $0 < x, y, z \le 9$

$\therefore$ Number of solutions = coefficient of $x^{20}$ in $\left(x + x^2 + ...x^9\right)^3$

= coefficient of $x^{17}$ in $\left(1 + x + ...x^8\right)^3$

= coefficient of $x^{17}$ in $\left(1 - x^9\right)^3 (1 - x)^{-3}$

$= 1 \times 19_{C_2} - 3 \times 10_{C_2}$

Required probability = $\dfrac{19_{C_2} - 3 \times 10_{C_2}}{19_{C_2}} = \dfrac{19 \times 18 - 3 \times 10 \times 9}{19 \times 18}$

$P = \dfrac{4}{19} \Rightarrow 19P = 4$.

88. 00008.00

Sol. Distance between the foci

$2ac = 4\sqrt{2}$

$ac = 2\sqrt{2}$

Product of distances of foci from tangent = $b^2$

$2 \times 6 = b^2$

$b^2 = a^2 - a^2e^2$

$12 = a^2 - 8$

$a^2 = 20$

$a = \sqrt{20}$

Major axis $2a = \sqrt{80}$

$[2a] = 8$.

89. 00001.00

Sol. $\dfrac{(\sin x)^{2y}}{(\cos x)^{\frac{y^2}{2}}} + \dfrac{(\cos x)^{2y}}{(\sin x)^{\frac{y^2}{2}}} \ge 2\sqrt{\dfrac{(\sin x \cos x)^{2y}}{(\cos x \sin x)^{\frac{y^2}{2}}}} = 2(\sin x \cos)^{y - \frac{y^2}{4}}$

$\Rightarrow \sin 2x \ge 2(\sin x \cos x)^{y - \frac{y^2}{4}}$

$\Rightarrow y - \dfrac{y^2}{4} \ge 1$

$\Rightarrow y = 2$ & $\sin x = \cos x$ $\left(\dfrac{\pi}{4}, 2\right)$

90. 00005.00

Sol. $x^2 = k(y - 4)$

$(x - 6)^2 = -ky$

writing tangent and comparing

$x_1 = 3$, $y_1 = 2$.